

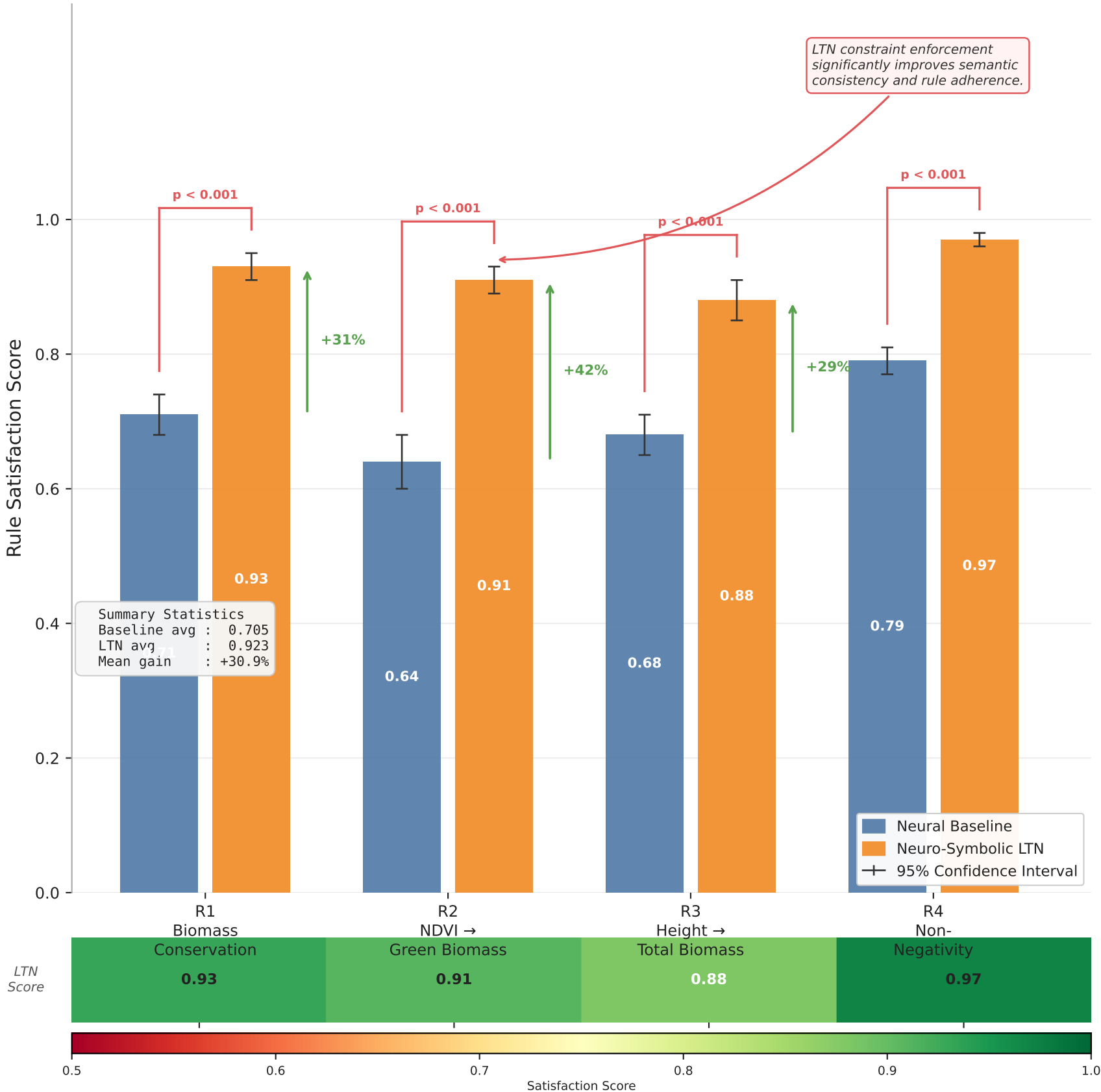
Neuro-Symbolic Knowledge Enforcement and Rule Satisfaction Analysis

Figure 4 — Logical Tensor Networks enforce domain-grounded agricultural constraints, improving semantic consistency beyond purely neural prediction.

(a)

Quantitative Rule Satisfaction Analysis

LTN constraint enforcement significantly raises all domain rule scores above neural baseline ($p < 0.001$)



(b)

Neuro-Symbolic Architecture

